Lecture 2:
Key Concepts and Terms

Understand the idea. Work through an example.

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Start with the question, then the calculation

1. **Understand the idea.** What is this number trying to measure?
2. **Follow the example.** Small numbers, one step at a time, with a calculator.
3. **Explain the answer.** What does it tell an investor or a manager?

Lecture 2 uses more symbols than Lecture 1. The first part here uses everyday language and small round numbers. The calculation reference at the back translates the symbols into words and repeats each formula with a short example.

Remember: This is a learning sequence, not a change to assessment requirements. Use the full lecture, the practice questions and the required reading too.

A return is a percentage of what you put in

The idea: A **return** states the gain as a share of the amount invested, so investments of different sizes can be compared.

Example: you pay £100 for a share and later sell it for £110

1. Money return: £110 − £100 = £10.
2. Percentage return: 10 ÷ 100 = 0.10, which is 10%.

What this means: A £10 gain on £100 and a £100 gain on £1,000 are the same 10% return.

Remember: Dividends and any other payments received count towards the return, not only the change in price.

Formula and notation reference →

Returns over different periods need a common yardstick

The idea: A six-month return and a one-year return cannot be compared directly. Convert both to an **annual equivalent rate (AER)**, as for these two investments, each turning £100 into £110.

Investment	Held for	Return over that period	Annual equivalent
A	6 months	10%	$1.10 \times 1.10 - 1 = 21\%$
B	1 year	10%	10%

What this means: A's six months of growth happens twice in a year, so A wins even though both pay £110.

Remember: Convert first, compare second: 300% over twenty years is only about 7% a year.

Formula and notation reference →

Expected return: average the past, or weight the scenarios

The idea: The **expected return** is our single best estimate of next period's return. There are two usual routes to it.

1. **From history:** add the returns that actually happened and divide by how many there are. Returns of 10%, -6% and 8% give $(10 - 6 + 8) \div 3 = 4\%$.
2. **From scenarios:** weight each possible return by its probability. A boom paying 30% and a slump paying -10%, equally likely, give $0.5 \times 30 + 0.5 \times (-10) = 10\%$.

What this means: In the lecture, five months of Barrick Gold returns average 3.1% a month. Five observations is a very short history to lean on.

Remember: The expected return is an average. It need not be a return you can actually receive in any single year: 10% here is neither 30% nor -10%.

Formula and notation reference →

Risk is the spread around the expected return

The idea: Two investments can share an expected return and still differ completely in how far the outcome can stray from it.

	Boom (50%)	Slump (50%)	Expected return
Share A	+30%	−10%	10%
Deposit account	+10%	+10%	10%

What this means: Share A strays 20 percentage points either side of its expectation. The deposit account never strays at all. No investor would treat the two as the same.

Remember: Risk counts surprises in both directions. A return of +30% is exactly as far from the expectation as −10% is.

Variance averages the squared distances

The idea: Variance measures the spread: take each outcome's distance from the expected return, square it, then average the squares.

Example: Share A, expected return 10%

1. Distances: $30 - 10 = +20$ points and $-10 - 10 = -20$ points.
2. As decimals, squared: $0.20^2 = 0.04$ and $(-0.20)^2 = 0.04$.
3. Average them: $0.5 \times 0.04 + 0.5 \times 0.04 = 0.04$.

What this means: Squaring removes the signs, so surprises in both directions add to the measure, and large surprises count far more than small ones.

Remember: Variance is in squared units, which is why it looks so small. The lecture's monthly Barrick Gold variance is 0.02473.

Formula and notation reference →



Standard deviation puts risk back into percentage points

The idea: The **standard deviation** is the square root of the variance, which returns the measure to the units of the return itself.

Example: Share A again

1. Variance: 0.04.
2. Square root: $\sqrt{0.04} = 0.20$, which is 20%.

What this means: "An expected return of 10% with a standard deviation of 20%" can be read straight off. A variance of 0.04 cannot.

Remember: The lecture's Barrick Gold figures are a variance of 0.02473 and a standard deviation of 15.7% a month. Risk is always quoted as the standard deviation.

Formula and notation reference →

The bell curve: one and two standard deviations

The idea: If returns follow a **normal distribution**, the expected return and the standard deviation together describe the whole picture.

Example: an expected return of 10% with a standard deviation of 20%

Range	Calculation	Roughly how often
One standard deviation	10 ± 20: -10% to $+30\%$	about two years in three
Two standard deviations	10 ± 40: -30% to $+50\%$	about nineteen years in twenty

What this means: A wider curve means a larger standard deviation. Doubling the standard deviation doubles the width of both ranges.

Remember: This is a model of returns, not a law. Real markets produce extreme years rather more often than the bell curve suggests.

A portfolio's return is a weighted average

The idea: A **portfolio** is a set of holdings. Its **weight** w on a share is the fraction of your money invested in it, and the weights add to 1.

Example: £600 in Share A, £400 in Share B

1. Weights: $600 \div 1{,}000 = 0.6$ and $400 \div 1{,}000 = 0.4$.
2. A is expected to return 10%, B to return 5%.
3. Portfolio: $0.6 \times 10 + 0.4 \times 5 = 6 + 2 = 8\%$.

What this means: The portfolio's expected return always sits between the two, closer to whichever holds more of your money.

Remember: This much is ordinary arithmetic. Combining the two risks is not.

Formula and notation reference →

A portfolio's risk is not a weighted average

The idea: Shares do not all fall on the same day. When one is above its expectation while another is below, the two partly cancel.

	Boom (50%)	Slump (50%)	Standard deviation
Share A	+30%	−10%	20%
Share B	−10%	+30%	20%
Half in each	+10%	+10%	0%

What this means: Averaging the two risks would give 20%. The portfolio's actual risk is zero: this is **diversification**.

Remember: Expected returns average. Risks do not: the average leaves out how the two shares move together.

Formula and notation reference →

Covariance: do the two move together?

The idea: Covariance multiplies the two shares' distances from their own expected returns, then averages those products. Here are Shares A and B, each expected to return 10%.

	A's distance	B's distance	Product
Boom	$+0.20$	-0.20	-0.04
Slump	-0.20	$+0.20$	-0.04

What this means: Both products are negative, because one share is always above its expectation while the other is below. Their average, -0.04, is the covariance.

Remember: That negative covariance is why the portfolio's risk vanished. A positive one means the two tend to land on the same side of their expectations.

Formula and notation reference →

Correlation rescales covariance onto a fixed scale

The idea: Correlation divides the covariance by the two standard deviations, which puts the answer between -1 and $+1$: for A and B, $-0.04 \div (0.20 \times 0.20) = -1$.

Correlation	What it means	Diversification
$+1$	they always move together	none: risk is the weighted average
0	no relationship	useful
-1	they always move opposite	greatest: risk can reach zero

Remember: Real shares sit between the extremes. Over 2/1985–8/2026 Wal-Mart and Honeywell returns correlated 0.25187, so mixing them removes some risk but never all of it.

The opportunity set: every mix is a point

The idea: Change the weights and you get a different portfolio, with its own risk and its own expected return. Plot risk across and expected return up, and the mixes trace a curve: the **opportunity set**.

- Putting all your money in one share puts you at that share's own point.
- Every mix in between bulges to the left of the straight line joining the two, because the correlation is below $+1$.
- The leftmost point is the **minimum-variance portfolio**. For the lecture's two shares it is 62% Wal-Mart and 38% Honeywell, with a standard deviation of 19.18% and an expected return of 15.68%.

What this means: You cannot reach a point above the curve. That would need higher returns, lower risks or lower correlations than the shares actually have.

Formula and notation reference →

The efficient frontier is the upper edge

The idea: Portfolios below the minimum-variance point are never worth holding: directly above each one sits a portfolio with the same risk and a higher expected return. The part of the curve above that point is the **efficient frontier**.

- **Efficient** means the highest expected return available at that level of risk.
- Moving further up the frontier always costs more risk. There is no free upgrade.
- Where an investor sits on it depends on how much risk they are willing to carry.

What this means: The frontier does not tell you which portfolio to hold. It tells you which ones to rule out.

Remember: Efficient is not the same as best. Any point on the frontier can be the right choice for some investor.

Formula and notation reference →

Minimum variance and optimal are different portfolios

The idea: The **minimum-variance** portfolio carries the lowest risk available. The **optimal** portfolio earns the most expected return per unit of risk. They are rarely the same mix.

Wal-Mart and Honeywell	In Wal-Mart	Standard deviation	Expected return
Minimum variance	62%	19.18%	15.68%
Optimal	70%	19.33%	15.92%

What this means: Moving from the first mix to the second raises the expected return from 15.68% to 15.92% for a little more risk. Whether that trade is worth making is a judgement about the investor, not about the shares.

Remember: “Lowest risk” and “best value for risk” are two different questions. Read which one you are being asked.

A short sale makes a weight negative

The idea: To **short** a share you borrow it, sell it now and buy it back later to return it, so you gain if the price falls. The proceeds go into your other holding, whose weight then passes 100%.

Example: the lecture's two shares

1. Short Honeywell: weight -32%.
2. Put everything, including the proceeds, into Wal-Mart: weight 132%.
3. The weights still add to 100%: expected return 17.85%, standard deviation 28.66%.

What this means: Short sales extend the curve beyond both end points. They raise expected return and risk together.

Remember: Weights may be negative, but they always add to one. A short position can lose without limit, because the buy-back price has no ceiling.



More shares push the frontier to the left

The idea: Each share added brings its own risk, but also more chances for movements to offset one another. Adding Agnico Eagle, Citigroup and Coca-Cola to the two shares shifts the whole curve left: less risk at the same expected return.

Minimum-variance portfolio	Standard deviation	Expected return
Two shares	19.18%	15.68%
Five shares	15.89%	15.72%

What this means: Three more shares cut the lowest achievable risk by over three percentage points, and the expected return does not fall to pay for it.

Remember: The gain is largest for the first few shares added, and it does not continue without limit.

The floor: market risk cannot be diversified away

The idea: Total risk = firm-specific risk + market risk. Diversification removes the first, because the risks specific to one firm average away, but not the second.

- **Firm-specific risk:** a failed product, a fire, a lawsuit. This averages away across many holdings.
- **Market risk:** interest rates, inflation, recession. This hits the whole portfolio together.

What this means: Among the lecture's five shares the average variance of one share is 0.1053, the average covariance between two 0.0156. Portfolio variance falls towards that covariance and stops.

Remember: This floor is why the second half of the lecture pays investors for market risk alone.

Formula and notation reference →

Adding a risk-free asset makes the line straight

The idea: A **risk-free asset** pays a known return with no variation; a short government bill is the usual stand-in. Because it never moves, it has no covariance with anything.

Example: a risk-free rate of 4%, and a risky portfolio at 12% with a 20% standard deviation

1. Half in each: expected return $0.5 \times 4 + 0.5 \times 12 = 8\%$.
2. Risk: $0.5 \times 20 = 10\%$. Only the risky half contributes any.

What this means: Mixing with the risk-free asset scales risk and the extra return by the same fraction, so every mix lies on a straight line from the risk-free point to the risky one.

Remember: Putting more than 100% into the risky portfolio means borrowing at the risk-free rate, which extends the same straight line further out.

Formula and notation reference →

The tangent portfolio gives the steepest line

The idea: Every risky portfolio offers a straight line out of the risk-free point. The steepest line is the best, because it pays the most extra return for each unit of risk taken. That slope is the **Sharpe ratio**.

Five shares, $r_f = 4\%$	Expected return	Standard deviation	Sharpe ratio
Tangent portfolio	16.17%	16.19%	$(16.17 - 4)/16.19 = 0.75$
Below the frontier	13.77%	20.81%	$(13.77 - 4)/20.81 = 0.47$

What this means: The second portfolio lies below the minimum-variance point, so its line out of the risk-free point is flatter: less reward for each unit of risk.

Remember: No risky portfolio has a higher Sharpe ratio than the tangent one. When it is the whole market, this line is the **capital market line**.

Beta measures how a share moves with the market

The idea: Beta (β) measures a share's sensitivity to the market: how far it typically moves when the market moves 1%.

Share	β	If the market rises 1%
Agnico Eagle	0.27	it rises about 0.27%
Coca-Cola	0.63	it rises about 0.63%
Wal-Mart	0.72	it rises about 0.72%
Honeywell	1.10	it rises about 1.10%
Citigroup	1.70	it rises about 1.70%
The market itself	1.00	it rises 1%, by definition

Remember: Beta is estimated over a chosen window and can move: Agnico Eagle's was -0.66 over 1/2011–1/2016 and 0.65 over 8/2021–8/2026.

Formula and notation reference →

The CAPM turns beta into a required return

The idea: The **capital asset pricing model (CAPM)** pays investors only for the risk they cannot diversify away: the required return is the risk-free rate plus beta times the market's risk premium.

1. A share with $\beta = 1.5$ carries 1.5 times the market's risk, so with a market risk premium of 6% it earns $1.5 \times 6 = 9\%$.
2. Add the risk-free rate of 4%: $4 + 9 = 13\%$ is the required return.

What this means: Plotting required return against beta gives the **security market line**, from the risk-free rate at $\beta = 0$ through the market at $\beta = 1$.

Remember: That required return is what a project of this risk must earn to satisfy shareholders: the **cost of equity**.

Formula and notation reference →



Alpha is the distance from the line

The idea: Alpha (α) is the part of a return the CAPM does not account for: the gap between what a share actually returns and what its beta says it should.

Example: the share above, whose required return is 13%

- Expected to return 15%: alpha is $15 - 13 = +2$ percentage points, and it plots above the line.
- Expected to return 11%: alpha is -2 points, and it plots below.

What this means: If everyone agreed on the model and the numbers, investors would buy the first share until its price rose far enough to pull the expected return back down to the line. Under the CAPM's assumptions every alpha is zero.

Remember: An alpha estimated from past data is not a promise of future returns. It may simply reflect risk that a single beta does not capture.

What the CAPM is for, and what is said against it

The idea: The CAPM's job in this module is to supply a discount rate: the return investors require for the market risk a project carries.

- **Roll (1977):** the true market portfolio holds every asset, so it cannot be observed and the model cannot be tested.
- **Unstable betas:** estimates depend on the window, as Agnico Eagle's change of sign shows.
- **Mixed evidence:** Fama and MacBeth (1973) found beta mattered; Fama and French (2006) found the relation unreliable.

What this means: Surveyed firms still use the CAPM for the cost of equity more than any other method.

Remember: Use it, state its assumptions, and remember that the cash-flow forecasts usually matter more than the discount rate.

Keep these questions attached to the terms

Term	The question it answers
Expected return	What return should I expect on average?
Variance	How far do outcomes stray from that, in squared units?
Standard deviation	How far do they stray, in percentage points?
Covariance	Do these two tend to move in the same direction?
Correlation	How strongly, on a scale from -1 to $+1$?
Efficient frontier	Which portfolios give the most return for their risk?
Sharpe ratio	How much return above the risk-free rate, per unit of risk?
Beta	How much market risk does this share carry?
CAPM	What return do investors require for that market risk?
Alpha	How far is the actual return from the required one?

Check your understanding before using the formulas

1. Why can a portfolio of two risky shares be less risky than either share alone?
2. What does a correlation of -1 allow that a correlation of $+1$ does not?
3. Why is nothing below the minimum-variance portfolio worth holding?
4. Why does adding a risk-free asset turn the curve into a straight line?
5. Why does the CAPM reward beta rather than standard deviation?

Answer in words first. Then support each answer with one small example.

Remember: If an answer reaches for a formula before it reaches for a sentence, go back to the idea.



Reference: returns and annual equivalents

In words: a return is the gain divided by the amount invested. To compare holding periods, ask what annual rate would compound to the same total.

$$r = \frac{P_1 - P_0 + D_1}{P_0} \qquad AER = (1 + r)^{1/n} - 1$$

P_0: price paid; P_1: price received; D_1: income received meanwhile; r: return over the holding period; n: length of that period in years.

Example: £100 grows to £110 in six months, so $r = 0.10$ and $AER = (1.10)^2 - 1 = 0.21$, or 21%.

Remember: Every rate belongs to a period. Keep rate and period consistent, as in Lecture 1.

Reference: expected return

In words: average the past returns, or weight each possible return by its probability.

$$\bar{r} = \frac{1}{T}\sum_{t=1}^{T} r_t \qquad E[r] = \sum_{s=1}^{S} p_s\, r_s$$

r_t: the return realised in period t; T: number of periods; p_s: probability of scenario s; r_s: the return in that scenario. The probabilities add to 1.

The large sigma means "add these up": run the index from the first value to the last, and total the terms.

Example: returns of 10%, −6% and 8% give $\bar{r} = (10 - 6 + 8)/3 = 4\%$. Two equally likely scenarios of 30% and −10% give $E[r] = 0.5(30) + 0.5(-10) = 10\%$.

Remember: A bar over a letter means an average. $E[\,]$ means an expected value.

Reference: variance and standard deviation

In words: average the squared distances from the expected return, then take the square root.

$$\sigma^2 = \frac{1}{T-1}\sum_{t=1}^{T}(r_t - \bar{r})^2 \qquad \sigma^2 = \sum_{s=1}^{S} p_s(r_s - E[r])^2 \qquad \sigma = \sqrt{\sigma^2}$$

σ ("sigma"): standard deviation; σ^2: variance. The first formula uses history, the second uses scenarios.

Why $T-1$: with historical data the mean is itself estimated from the same returns, so one observation is used up. Scenario probabilities need no such correction.

Example: scenarios of 30% and -10%, each with probability 0.5 and an expected return of 10%, give $\sigma^2 = 0.5(0.20)^2 + 0.5(-0.20)^2 = 0.04$ and $\sigma = 0.20$, or 20%.

Reference: portfolio expected return

In words: multiply each holding's expected return by its share of the money, then add.

$$\bar{r}_p = \sum_{i=1}^{N} w_i \bar{r}_i \qquad \text{with} \qquad \sum_{i=1}^{N} w_i = 1$$

w_i: the fraction of the portfolio held in security i; $\bar{r}_i$: that security's expected return; N: how many securities there are. A negative weight is a short position.

Example: $w_1 = 0.6$ with $\bar{r}_1 = 10\%$, and $w_2 = 0.4$ with $\bar{r}_2 = 5\%$, give $\bar{r}_p = 0.6(10) + 0.4(5) = 8\%$.

Remember: This one really is a weighted average. Portfolio risk is not, which is why it needs its own page.

Reference: covariance and correlation

In words: covariance averages the product of the two distances from their own means. Correlation divides that by both standard deviations, which puts it on a fixed scale.

$$\sigma_{ij} = \frac{1}{T-1}\sum_{t=1}^{T}(r_{t,i} - \bar{r}_i)(r_{t,j} - \bar{r}_j) \qquad \rho_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j}$$

σ_{ij}: covariance between securities i and j; ρ_{ij} ("rho"): their correlation, always between -1 and $+1$.

Example: two equally likely scenarios with distances $(+0.20, -0.20)$ and $(-0.20, +0.20)$ give $\sigma_{ij} = -0.04$; with $\sigma_i = \sigma_j = 0.20$, $\rho_{ij} = -0.04/(0.20 \times 0.20) = -1$.

Remember: Rearranged, $\sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$. That is how correlation gets into the portfolio-risk formula.

Reference: portfolio risk with two assets

In words: add each holding's own contribution, then twice the cross term that carries the covariance.

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2$$

Example: $w_1 = w_2 = 0.5$ and $\sigma_1 = \sigma_2 = 0.20$. The first two terms give $0.01 + 0.01 = 0.02$ whatever the correlation; only the cross term changes.

ρ_{12}	Cross term	σ_p
$+1$	$+0.02$	20%
0	0	14.1%
-1	-0.02	0%

Remember: Only at $\rho_{12} = +1$ is portfolio risk the weighted average of the two risks. Below that it is always less.

Reference: the frontier and the minimum-variance mix

In words: with many assets, sum the covariance term over every pair, including each asset with itself. The frontier traces the lowest risk available at each expected return.

$$\sigma_p = \sqrt{\sum_{i=1}^{N}\sum_{j=1}^{N} w_i w_j \sigma_{ij}} \qquad w_1^{\min} = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$$

The double sum runs over all $N \times N$ pairs; where $i = j$ the covariance σ_{ii} is just the variance σ_i^2.

Example: $\sigma_1 = 20\%$, $\sigma_2 = 10\%$ and $\sigma_{12} = 0$ give $w_1^{\min} = 0.01/(0.04 + 0.01) = 0.20$: holding 20% in the riskier share leaves a standard deviation of 8.9%, below either share alone.

Remember: The minimum-variance portfolio is the left-hand tip; the efficient frontier is everything above it.

Reference: the risk-free asset and the Sharpe ratio

In words: mixing with a risk-free asset scales risk and extra return by the same fraction, so the combinations lie on a straight line whose slope is the Sharpe ratio.

$$\bar{r}_p = r_f + \frac{\bar{r}_2 - r_f}{\sigma_2}\,\sigma_p \qquad \text{Sharpe ratio} = \frac{\bar{r} - r_f}{\sigma}$$

r_f: the risk-free rate; $\bar{r}_2$ and σ_2: the risky portfolio's expected return and risk. Here $\sigma_p = w_2\sigma_2$, because the risk-free asset has no variance and no covariance.

Example: $r_f = 4\%$ with a risky portfolio at 12% and $\sigma_2 = 20\%$. Its Sharpe ratio is $(12-4)/20 = 0.40$. Half in each gives $\sigma_p = 10\%$ and $\bar{r}_p = 4 + 0.40(10) = 8\%$.

Remember: This line is the capital allocation line, and the capital market line when the risky portfolio is the whole market.

Reference: beta

In words: beta is a share's covariance with the market divided by the market's variance, which is also the slope of a regression of the share's excess return on the market's.

$$\beta_i = \frac{\sigma_{im}}{\sigma_m^2} = \rho_{im}\frac{\sigma_i}{\sigma_m} \qquad (r_i - r_f) = \alpha_i + \beta_i(r_m - r_f) + \varepsilon_i$$

σ_{im}: covariance with the market; σ_m: the market's standard deviation; ε_i ("epsilon"): the part of the return the market does not explain, whose expected value is zero.

Example: $\sigma_i = 30\%$, $\sigma_m = 15\%$ and $\rho_{im} = 0.5$ give $\beta_i = 0.5 \times 30/15 = 1.0$. A share can be twice as volatile as the market and still carry only average market risk.

Remember: Only the market-related part of a share's movement counts here. The rest diversifies away.

Reference: the CAPM and the security market line

In words: the required return is the risk-free rate plus beta multiplied by the market's risk premium.

$$\bar{r}_i = r_f + \beta_i(\bar{r}_m - r_f) \qquad \alpha_i = \bar{r}_i^{\text{actual}} - \left[r_f + \beta_i(\bar{r}_m - r_f)\right]$$

$\bar{r}_m - r_f$: the market risk premium, the extra return expected from the market over the risk-free rate. For a portfolio, $\beta_p = \sum_i w_i \beta_i$.

Example: $r_f = 4\%$, a market premium of 6% and $\beta_i = 1.5$ give $\bar{r}_i = 4 + 1.5(6) = 13\%$. At $\beta = 0$ the answer is 4%; at $\beta = 1$ it is the market's own expected return.

Remember: The security market line is this equation drawn out: beta across, required return up. Alpha is vertical distance from it.

Reference: splitting total risk

In words: a share's variance divides into a market part and a part belonging to the firm alone.

$$\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_{\varepsilon_i}^2$$

$\beta_i \sigma_m$: the market part, as a standard deviation; σ_{ε_i}: the firm-specific part, which diversification removes.

Example: $\beta_i = 1.5$ with $\sigma_m = 20\%$ gives a market part of $1.5 \times 20 = 30\%$. If the total standard deviation is 50%, the firm-specific part is $\sqrt{0.50^2 - 0.30^2} = 0.40$, or 40%.

In the lecture: Coca-Cola's total standard deviation is 19.90% a year, of which $\beta_i \sigma_m = 9.55\%$ is market risk. The rest is firm-specific.

Remember: The two parts add as variances, not as standard deviations, which is why the square root appears.